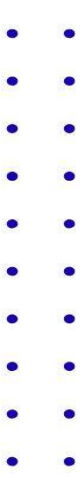


OBJECT-ORIENTED SYSTEM
ANALYSIS AND DESIGN

ESTATE AGENCY SYSTEM

AUTUMN 2024 – WINTER 2025



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Preface

North Yorkshire Property services is a business about with property sales. It has CEO, manager, representatives (part-time and full-time) and clients (buyer and seller). It has five branches. Each property has one owner. It is managed from a single branch.

So, they need property viewing system to manage its property sales operation and thus the stakeholders can communicate and work through PVS.



Fig 1. Estate agency system (freepik, 2026)

Task 1 – Elicitation of requirements

a) Stakeholders in PVS system

Chief executive officer (CEO)

CEO is main sponsor and project initiator of the property viewing system (PVS). They support business goals more than daily operational use. They give financial support for the project.

Branch managers

They have administrative and supervisory authority at each branch. They manage branches and representatives' records. They are five branches and so there may be five managers.

Representatives

They are primary operational users of the property viewing system (PVS) for managing clients , properties, offers and viewing appointments. They include part-time and full-time employees in each branch. Specific branch manager manages representatives from each branch.

Clients

They are external stakeholders who use PVS. They include buyers who browse and search properties and sellers are owners of properties who market them through the PVS.

b) Methods to gather system requirements

Interview

It is one-to-one or small-group discussion with prepared follow-up questions (Open questions). Branch managers, CEO and representatives will be interviewed.

Multiple points of view can be obtained and so interview provides detailed qualitative requirements for managing clients, properties, appointments and offers. Complex processes and non-functional requirements such as security can be understood clearly.

(Beaton, 2024)

Survey

Survey is structured questionnaires that are written in paper or online form. They usually include closed questions such as MCQ and scale rating and optional open questions. Representatives and clients will be surveyed through online or in paper form.

It is less-time consuming and appropriate for large groups to collect quantitative requirements. It is fast to collect and easy to compare data. It can prioritize common requirements such as property search criteria and consistency of branch processes.

(Motion, 2023)

c) Additional required information

1. Access control and permission

It is essential to identify role-based access for manager, representatives and clients to protect data integrity of NYPS.

2. Detailed information of full-time and part-time representatives

It is required for scheduling and managing workload and defining role and responsibilities clearly.

3. Detailed authentication methods and authorization

It is essential that only authorized users can access and modify about the system to protect sensitive data about NYPS.

4. Detailed appointment scheduling rules

It is needed to prevent double booking, check availability of appointments and show clear communication of stakeholders when booking.

5. Reporting features

It is required for monitoring, performance analysis and decision-making for manager.

(Struk, 2025)

Task 2 – Specification for requirements

a) Difference between functional and non-functional requirement

Functional requirement

It means what the system does and provides specific functions and services of system.

For example, in PVS system, clients can browse and search about price, type, district and numbers of bedroom. The representative can manage property, client, offer and appointment.

Non-functional requirement

It means how the system behaves and shows a condition under which system operates such as security, performance and usability.

For example, in PVS system, the system has fast response (performance) and safe security for protecting user information and data.

(GeeksforGeeks, 2020)

b) Functional requirements

- 1) Staffs are allowed to edit, add and delete about properties, offers, appointments and clients.
- 2) Managers can manage about representatives' and branches' data (add, edit and delete).
- 3) Clients are allowed to search and browse properties by type, price, number of bedrooms and district.

Additional functional requirements

1) Registration function (user login and password)

It is required for authentication and authorization for improving security of system.

2) Appointment making, scheduling and notification function

It is essential for clear communication between stakeholders and prevent conflict within system.

3) Role-based access control function

It is required for protecting integrity of data within NYPS as each user can only access information with its specific role.

4) Reporting features' function

It is needed for manager to monitor, analyze system and then make decisions.

c) Non-functional requirements

1) The system has fast response (performance).

For example, when a user clicks “View” Button in PVS, the system immediately show the available properties in 2.5 second.

2) The system protects personal data securely.

For example, when users want to log in into system, they must first register with correct information.

Additional non-functional requirements

1) Usability

The system must be easy to use for the clients (user-friendly). It should have clear user interface design and clear labels.

2) Maintainability

The system must be easy to update and fix when required. Updates should be released regularly.

3) Availability

System can be accessed at any time (99.9% uptime) with minimal downtime. So, it should have backup data storage.

4) Portability

The system can be opened in different devices without compatibility issues. It should be opened in all browsers that exist in all different devices.

(altexsoft, 2023)

Task 3 – Use case model

a) Purpose of use case model

It shows system functional requirements from users' point of view. It uses association such as include, extend and generalization. It describes what system should do before designing and coding. (Paradigm, 2023)

It supports communication, clearly understanding and can check requirements in early stage. For example, in PVS, roles such as manager, representative and client and key functions such as making appointment are clearly defined.

b) Actors in PVS

1) Client

Clients make sign-in, register, sign-out, browse and search properties. They include seller and buyer.

2) Seller

Sellers markets their properties through the property viewing system.

3) Buyer

Buyers make appointments for viewing properties with representatives.

4) Employee

Employees make sign-in, register, and sign-out. They include manager and representative.

5) Manager

Manager manages employees and branches by adding, editing and deleting. Each branch has its specific employees.

6) Representative

Representatives manage properties, offers, appointments and clients by adding, amending and deleting. They include part-time and full-time staffs.

7) Part time staff

Part time staffs are the employees who work part time for PVS in each branch.

8) Full time staff

Full time staffs are the employees who work full time for PVS in each branch.

c) Use case diagram for Property Viewing System (PVS)

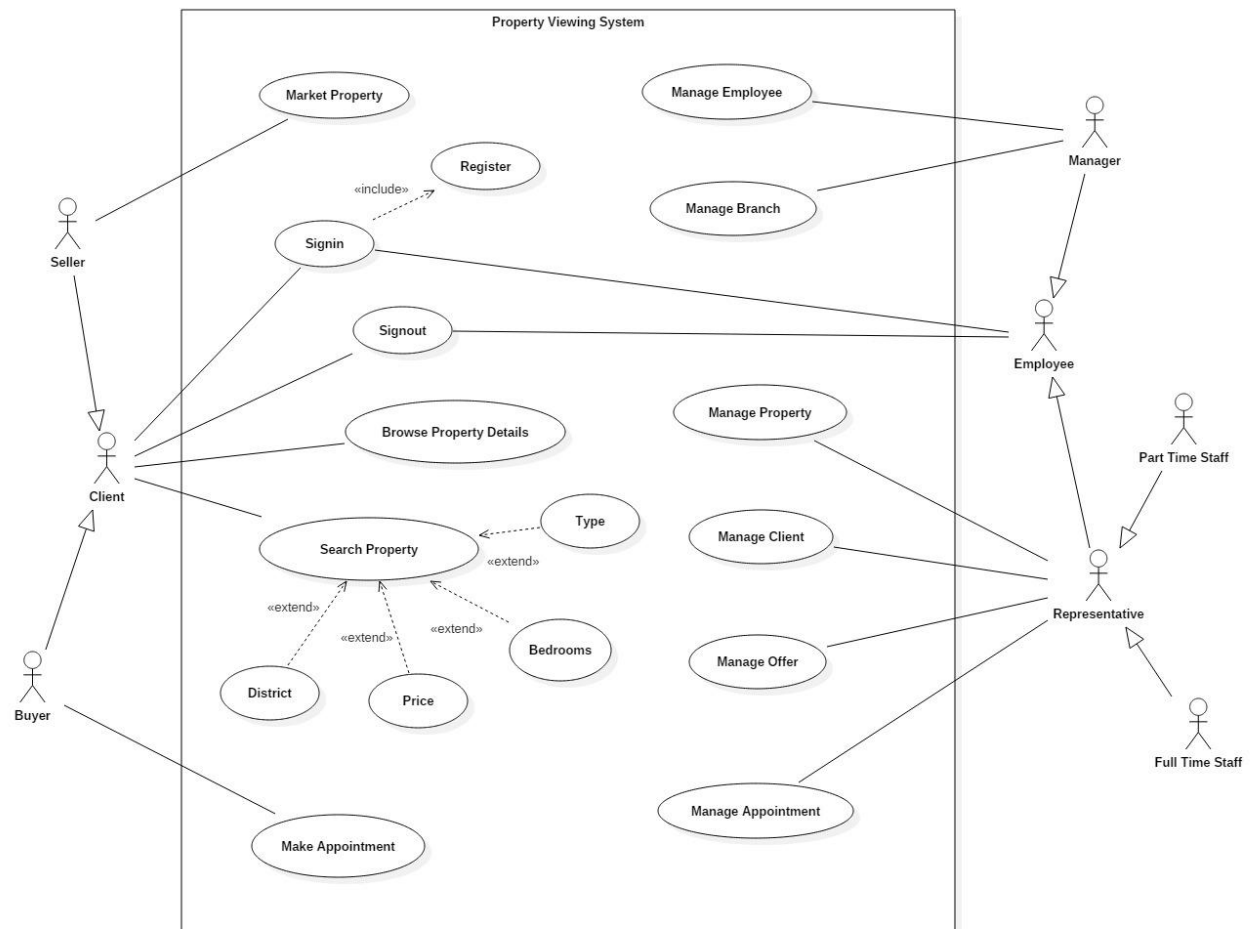


Fig 3.1 – Use case diagram of PVS

d) Scenarios for use case Make appointment

Scenario from buyer view

When a buyer wants to buy a property, he needs to check properties in PVS. So, he must first register and login. Then, he will see home page and search about the properties with his desired characteristics. When he viewed all the properties and found the wanted one, he will request appointment with desired date and time to view property in real condition (not online). Then, he needs to wait for confirmation from the system. That property is added to the system by seller and confirmed by the representative.

Scenario from representative view

When a new appointment request is received from buyer, system notifies it to a representative. Then, representative checks into database that the property is available or not and also the date and time are available or not. When property, date and time are available, representative confirms appointment and sends it to buyer through system. If buyer wants to change date and time, representative checks again into database and update appointment. If the buyer wants to cancel appointment, the representative will cancel it and update system.

e) Documentation about Use Case for Make appointment

Document	Description
Task Name	Create an appointment
Associated Actors	Representative, Buyer, Seller
Starting Conditions	Buyer found wanted property and selected it when viewing properties in PVS.
Process Flow	1. System responds with new screen which appears with detailed information of property for making appointment. 2. The buyer needs to enter his name, ID, property ID, desired date and time to view it outside. 3. The system checks that property is available or not. 4. If property is not available, system displays an appropriate message. 5. If property is available, system asks buyer that it is confirmed to request appointment. 6. Representative confirms or cancels appointment. 7. The system forwards request to representative. 8. Representative receives notification of requested appointment. 9. Representative checks it into database that the property, date and time are available or not. 10. After confirming that all are available, representative sends confirmation letter to buyer through system.
Completion Outcomes	If appointment was confirmed, it will be recorded into system. If it is not confirmed, the system does not make changes.
System Constraints	To make appointment, both the buyer and seller of the property must be registered. Buyer must be logged into system when making appointment.

Task 4 – Sequence diagram

a) Purpose of sequence diagram

It shows time-ordered interactions between system components. It contains actors, components, lifelines (time flows from top to bottom), messages (horizontal arrows show who calls whom and in what order) and activation bar that indicates the object is actively processing.

It shows step by step tracing from request to final response. It clearly defines which components performs what action and so complex processes can be clearly seen into simple interactions. Missing steps and logic gaps of the system can be found early. Visual reference can be shared for both technical and non-technical stakeholders. It improves understanding, collaboration and documentation of software development.

(Innovate, 2024)

b) Sequence diagram for property viewing system (PVS)

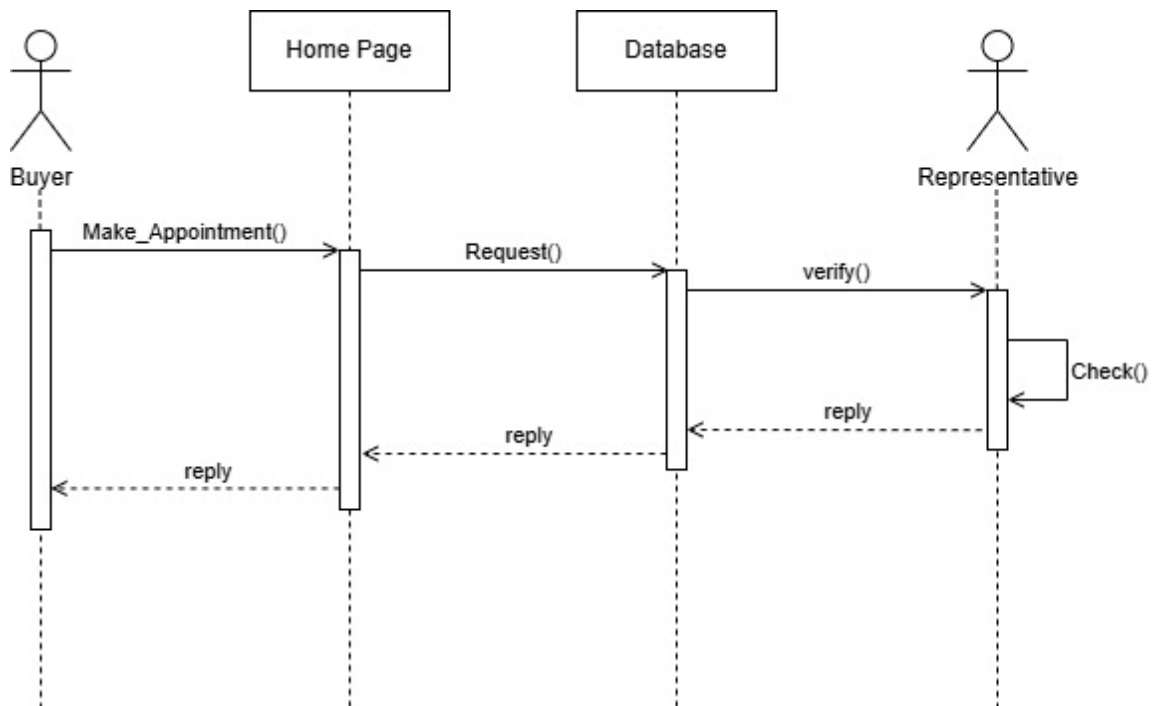


Fig 4.1 – Sequence diagram of PVS

Task 5 – Class diagram

a) Class diagram of property viewing system (PVS)

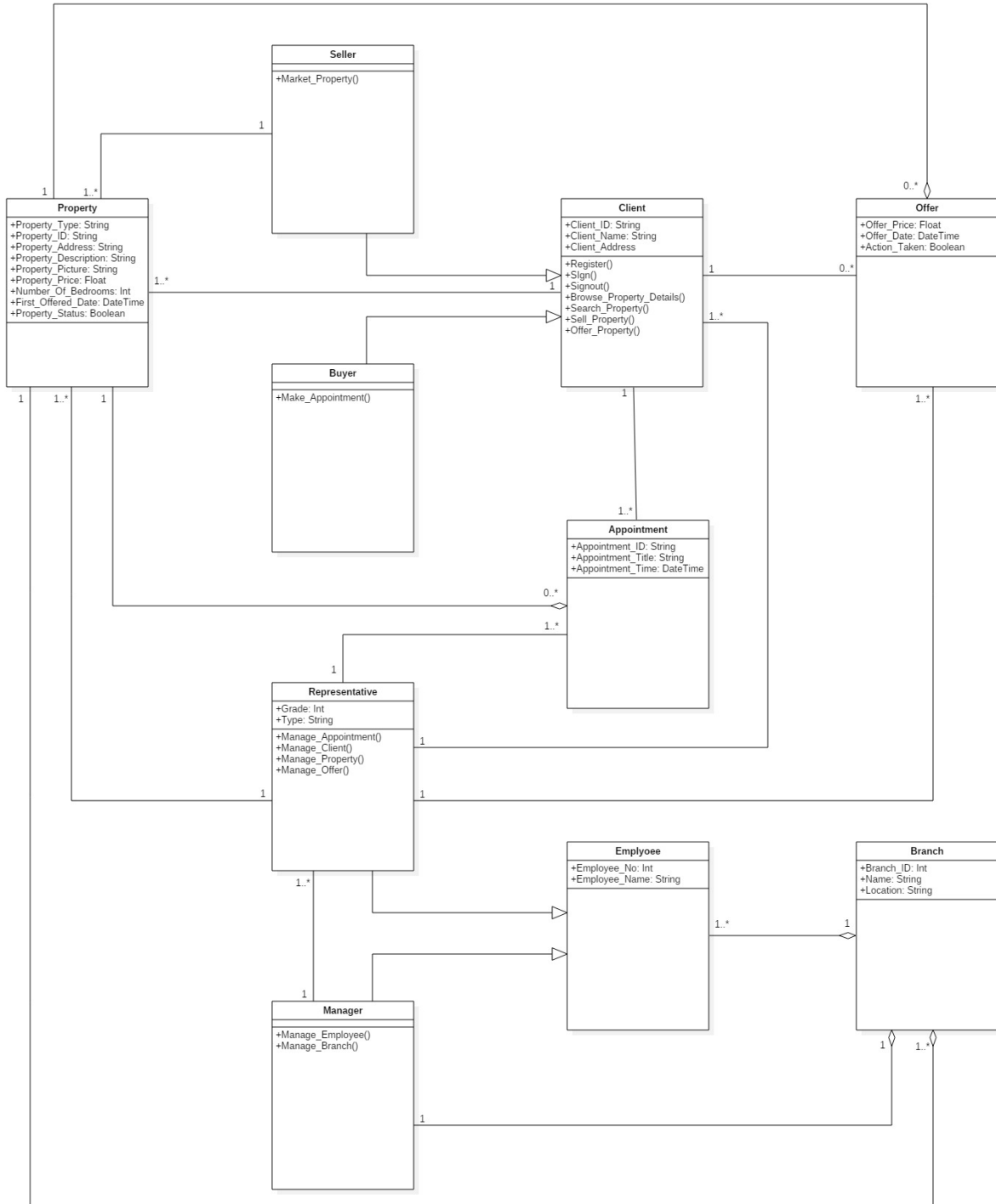


Fig 5.1 – Class diagram of PVS

b) Justification for each class

People, things, category and concept can be defined as class according to the system requirements.

Client class (include seller and buyer)

Client is a type of people that include seller and buyer. The scenario also gives information about clients such as names and address. They can sell, view and offer properties. So, Client is made as parent class and Seller and Buyer are linked to it with generalization. Seller and Buyer are child classes of Client. Seller can market property and buyer can make appointment for property.

Employee class (include representative and manager)

Employee is a type of people that include representative and manager. The scenario also gives information about employees such as names and number and about representative such as grade and type. They are the part of the branch. So, Employee is made as parent class and Representative and Manager are linked to it with generalization. Representative and Manager are child classes of Employee. Representative can manage appointment, client, property and offer. Manager can manage employee and branch.

Branch class

Branch is a type of category. The scenario also gives information about branch such as names, locations and staff. Employees are a part of branch and it is also linked with property with association.

Property class

Property is a type of things. The scenario also gives information about property such as description, address, pictures, type, price, number of bedrooms, first offered date and property status.

Appointment class

Appointment is a type of concept. The scenario also gives information about appointment such as representative can manage appointment with client for viewing property.

Offer class

Offer is a type of concept. The scenario also gives information about offer such as date, offer price and action taken. Client can make offer about property with representative.

Conclusion

The stakeholders and the function and non-functional requirements of the PVS are clearly defined. The methods of gathering requirements such as interview, survey are clearly explained. Sequence diagram, class diagram and use case diagram are clearly defined to build the Property Viewing System with minimal errors.

-----End of the Assignment-----

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